Comprehensive Network Infrastructure for a Mid-Sized Company

By Lan Anh Chung

Company Needs

- Support core services: web hosting, email, VoIP, file sharing, and database access
- Enable secure remote access for home and mobile employees
- Provide reliable, high-performance connectivity as headcount and services grow
- Maintain strong security without slowing down day-to-day operations

Approach

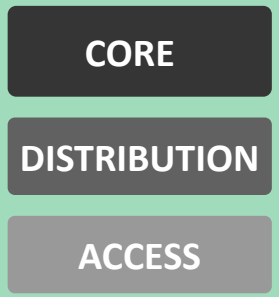
- Three-tier core / distribution / access switching architecture
- Classless (VLSM) IPv4 addressing with VLAN-based segmentation
- Layered, defense-in-depth security with a structured troubleshooting method
- Centrally managed wireless and MFA-protected remote access

Network Architecture

- **Tiered Switching Design**
  Three Tier Architecture
- **IP Addressing**
  Classless
- **Subnet**
  VLSM Addressing Plan

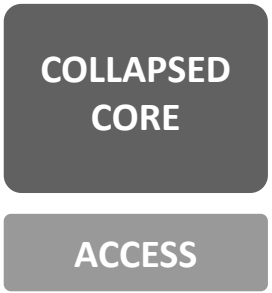
Three-Tier vs. Two-Tier Architecture

Three-Tier



Pros: Maximum Scalability and performance; security policies do not slow down core traffic; high redundancy.
Cons: Higher initial hardware cost and complex wiring.

Two-Tier



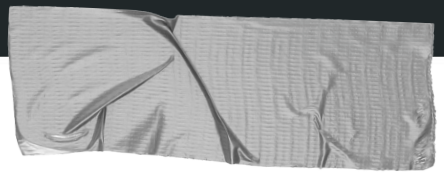
Pros: Lower hardware costs, simpler to set up and manage.
Cons: Limited scalability; a bottleneck occurs when routing, security policies (ACLs), and high-speed data flow run on the same physical devices.

Comparing IP Addressing & Subnetting Approaches

Classful / Flat, Fixed-Size Subnets	**Classless VLSM, Hierarchical (Chosen)**
• Fixed subnet mask (e.g., all /24) regardless of how many hosts a segment needs	• Variable-length masks size each subnet to its actual need (e.g., /23 for users, /24 for servers)
• Simple to plan and document by hand	• Hierarchical, function-based VLANs isolate server, user, voice, and wireless traffic
• Wastes address space on small segments; runs out quickly on large ones	• Efficient use of address space with room reserved for growth
• Poor traffic isolation — larger broadcast domains slow performance as the network grows	• Smaller broadcast domains and clear ACL/QoS boundaries improve traffic management

VLSM Addressing Plan

VLAN	Purpose	Subnet	Hosts
10	Servers (Web/DB/File)	10.10.10.0/24	254
20	Workstations	10.10.20.0/23	510
30	VoIP Phones	10.10.30.0/23	510
40	Wireless (Corp/Guest)	10.10.40.0/24	254
99	Remote VPN Pool	10.10.99.0/24	254



Network Interfaces and Devices

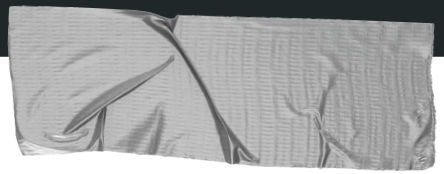
- **Ethernet-Switching-Features**
  VLANs, Voice VLAN, Port Security
- **Types of network interfaces**
  Copper vs. Fiber

Core Ethernet Switching Features

- **VLANs (IEEE 802.1Q):** Strict logical traffic isolation across all 5 segments.
- **Voice VLANs:** Automatically prioritizes VoIP traffic at the port level using LLDP-MED.
- **Port Security:** Enforces static or "Sticky" MAC address binding on Access ports to prevent rogue device attachments.

Interface Type

Interface Type	Primary Use Case	Pros	Cons / Trade-offs
Copper	Workstations, VoIP, Access APs	Cost-effective, supports **PoE+** up to 100m	Prone to EMI, lower maximum bandwidth
Fiber	Core Uplinks, Server Datacenter	High-speed, immune to EMI, future-proof	High cost for transceivers and deployment



Security and Troubleshooting

- **Security Hardening**
  Layered Security Measures.
- **Troubleshooting**
  Structured OSI Model Methodology

Network Hardening Techniques

Security Solution	Pros	Cons / Trade-offs
Perimeter Firewall + ACLs *(Stateful Edge Filter + Router Rules)*	• Well-understood & industry standard. • Enforces **least-privilege** internally (e.g., blocks users from database ports).	• **Static rules only**: No content inspection. • Large rule sets are hard to audit and maintain over time.
Next-Gen Firewall (NGFW) *(Deep Packet Inspection + IPS)*	• **Deep Packet Inspection (DPI)**. • Catches malicious payloads hidden inside "allowed" ports (e.g., HTTPS).	• Higher hardware and license costs. • Adds processing overhead and requires constant fine-tuning.

Authentication Methods

Security Solution	Pros	Cons / Trade-offs
Option 1: Local / Shared Credentials *(Stored on each device individually)*	• Simple to set up. • No dependency on a central server (always works).	• **Does not scale**: Must manage passwords per device. • No central audit trail. • One compromised password risks every device.
Option 2: Centralized AAA + MFA *(RADIUS/TACACS+ with Multi-Factor)*	• Central control and full logging. • **Instant access revocation** for individual users. • **MFA** blocks stolen password attacks.	• Creates a single point of failure (requires server redundancy). • More complex initial deployment.

Trouble Shooting

Troubleshooting Approach: Structured **OSI Model Methodology**

Network Diagnostics Comparison

Diagnostic Tool	Target Layer	Primary Strength	Weakness / Trade-off
Ping & Traceroute	Layer 3 (Network)	Instant verification of end-to-end IP reachability and path failures.	Basic feedback; cannot diagnose higher-layer protocol errors.
Nmap (Port Scanner)	Layer 4 (Transport)	Validates firewall rules and open services across network hosts.	Passive scans miss deeper issues; can trigger security alerts.
Wireshark (Packet Analyzer)	Layers 2–7 (Data to App)	Deep packet inspection; isolates complex VoIP jitter or application lag.	High complexity; requires deep protocol knowledge to interpret data.

Wireless and Remote Access

- **Wireless-Deployment**
  WPA3
- **Remote Access**
  VPN

Wireless Deployment

	WPA2-Enterprise	WPA3-Enterprise (Chosen)
Encryption	AES-CCMP	AES-GCMP (192-bit option)
Key exchange	4-way handshake	SAE ('Dragonfly') — resists offline dictionary attacks
Forward secrecy	No	Yes — past traffic stays protected if a password leaks
Open network protection	None	Enhanced Open (OWE) encrypts even open guest Wi-Fi

Remote Access

Feature / Criteria	Client-to-Site VPN *(Chosen)*	SSH Tunneling + RDP
Protocol Foundation	IPsec / WireGuard (Layer 3)	SSH / Remote Desktop Protocol (Layer 7/4)
Security Architecture	**Maximum**	**Moderate**
Scalability	**Excellent:** Thousands of concurrent sessions handled by a single network appliance.	**Poor:** Binds office hardware resources 1:1; high administrative overhead.
User Experience	**Seamless:** Feels exactly like working on the local office network.	**Laggy:** High sensitivity to latency; visual lag during heavy application use.

Conclusion

- **Future-Proof Scalability**
  Three-Tier Architecture, non-continuous VLSM plan
- **Robust Defense-in-Depth Security**
  IEEE 802.1Q VLANs and Extended ACLs,
  WPA3-Enterprise and MFA-protected VPNs
- **Operational Efficiency**
  Voice VLANs, OSI-model troubleshooting methodology

Thank you!